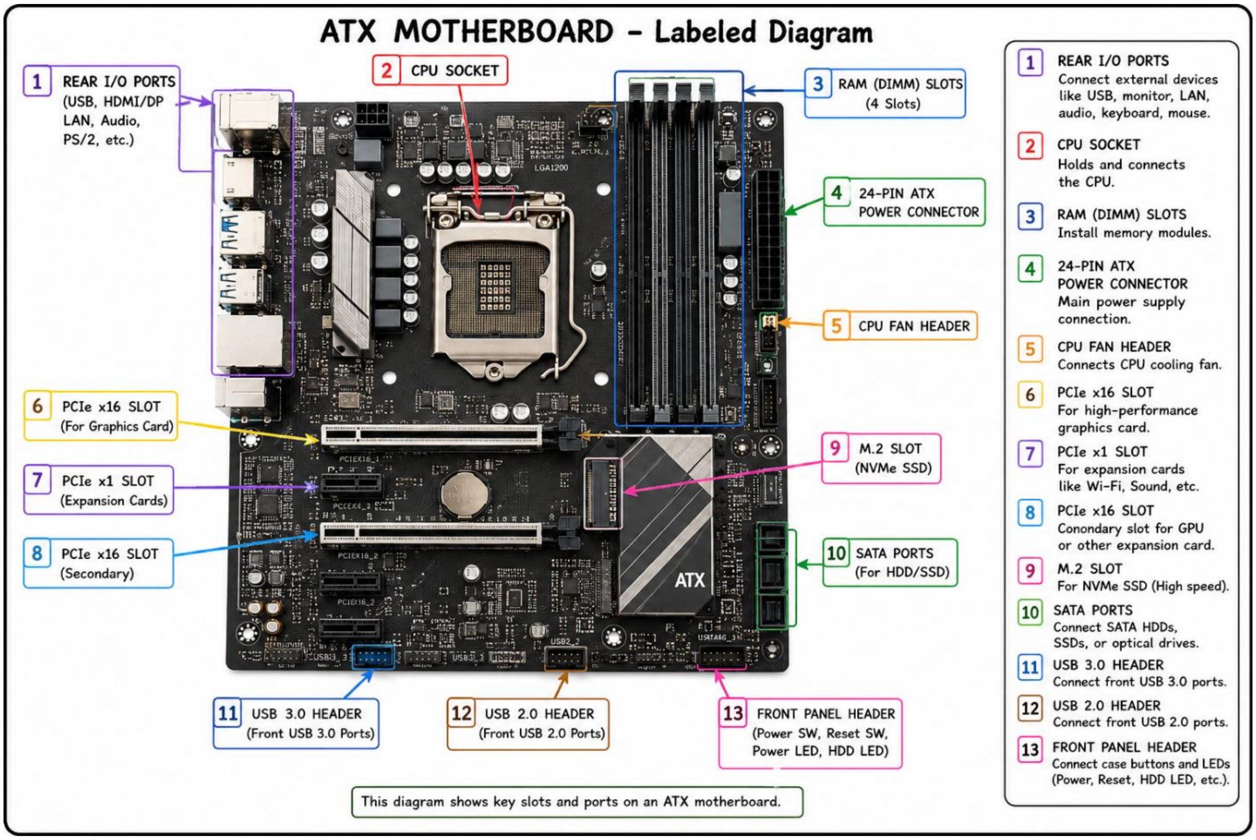
Q-1.Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).

ANSWER:-



Q-2 .Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

ANSWER:-

1. LGA1151 (Intel):

LGA1151 is an Intel desktop CPU socket that supports 6th and 7th Generation Intel Core processors. It is commonly used in gaming and office desktop computers. **Compatible CPU Model: ** Intel Core i7-7700K.

2. AM4 (AMD):

AM4 is a popular AMD desktop CPU socket that supports a wide range of Ryzen processors from the 1000 to 5000 series. It is known for its excellent performance, affordability, and long upgrade path. Compatible CPU Model: AMD Ryzen 5 5600X.

3. LGA1200 (Intel):

LGA1200 is an Intel CPU socket designed for 10th and 11th Generation Intel Core processors. It offers improved power delivery and performance for modern desktop PCs. **Compatible CPU Model: Intel Core i5-10400F.

Q- 3. Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

ANSWER:-

The **CPU (Central Processing Unit)** is the main processor of the computer. It reads instructions, performs calculations, and controls almost every task the computer does. You can think of it as the **"brain"** of the computer.

The **chipset** is a group of controller chips on the motherboard that manages communication between the CPU and other hardware, such as the RAM, storage (SSD/HDD), USB ports, graphics, and network devices. It acts like a **traffic manager**, making sure data moves to the correct component.

How they work together when you open Spotify

When you click the **Spotify** icon:

1. The **CPU** receives the command to open the app.
2. The **chipset** helps the CPU access the Spotify program files from the SSD or HDD and transfers the data to the RAM.
3. The **CPU** loads the program into memory and starts running its instructions.
4. The **chipset** continues to manage communication between the CPU, RAM, storage, audio hardware, and internet connection.
5. The **CPU** processes your actions (such as searching for a song or pressing Play), while the **chipset** ensures the required data moves quickly between the different hardware components so the music plays smoothly.

Q-4. Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labeling each one.

ANSWER:-